

Reproducibility materials

Frequency Saturation and Certified Computation of Relaxation Functionals

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Archive: `03_Reproducibility_Materials.zip`. The archive contains the exact inputs, frozen certificates, checker source, recorded seeds, and table-generation scripts supporting Section 7 of the manuscript. All central proofs are contained in the manuscript. A file manifest identifies the complete code and data snapshot by SHA-256 hashes.

Quick start. Extract the archive, open the resulting `Supplementary_Materials` directory, and execute

```
python3 run_all.py
```

Python 3.10 or later suffices for frozen verification; the replay uses only the standard library. A fresh run verifies the manifest and case inventory, runs the checkers in disposable working directories, and regenerates the numerical tables. The report is written to `replay_output/SUMMARY.json`. Optional candidate regeneration and its additional dependencies are described separately in `REGENERATION.txt`.

Evidence map.

Sections 3–4; Section 7.1; Table 1; Figure 1. Twenty-two certificates for full-frequency positive pairs, four polynomial reproduction maps, and twelve equal-effort design certificates. The frequency supremum is certified by an exact response identity and a rational enclosure of its unique maximizer. Figure 1 is drawn from the closed-form identity by the script `make_figure_response_difference.py` in the `figures` directory of the manuscript source, after a cross-check against the archived enclosures.

Proposition 4.6; Section 7.2. Sixteen zero-noise identification certificates, with positive spectra, identical original observations, and nonzero target differences.

Sections 5 and 6; Section 7.3 (Tables 2 and 3). Eighteen stopped-cost certificates and two executed stopping trajectories. The trajectories include six stage inputs and twelve continuous endpoint certificates.

Section 7.4 (Table 4). Six strict-feasibility anchors on quadratic guard grids, the relative-width checks, and table regeneration.

Replay accounting. The frozen suites comprise 80 certificate tasks and 154 named algebraic, mechanism, and invalid-input checks, including 53 intentional rejections. The twelve continuous endpoints occur inside the two trajectory tasks. These counts describe computational checks of the archived evidence. The two seeded trajectories exercise the stopping algorithm; the coverage argument is proved in Section 6.

Provenance and navigation. `SECTION_MAP.txt` gives exact paths, `INDEX.txt` lists the shipped files, and `MANIFEST_SHA256.json` records their hashes. Original suite names and receipts retain their provenance. `TRANSFER_NOTE.txt` explains the title change and the metadata updates for the present submission. Copyright and third-party dependency information appear in `RIGHTS_AND_DEPENDENCIES.txt`.